What Makes a Virginia Crash Deadly? A Multi-Method Analysis of 379,000 Crashes (2022–2024)

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Abstract

Traffic crashes killed over 40,000 people nationally in 2023, yet only a small share of all crashes are fatal. Understanding *which* crash characteristics raise the probability of a fatality — and by how much — informs both driver behavior and infrastructure investment. Using the public Virginia Open Data Portal release of 379,234 police-reported crashes from 2022–2024, we apply chi-square tests, multivariable logistic regression, classification trees (CART), and random forests to identify the strongest predictors of crash fatality. Three methods agree that **alcohol involvement, vulnerable road users (pedestrians, motorcyclists, cyclists), speed, and rural locations** are the dominant risk factors, with adjusted odds ratios ranging from 2 to 23. We also document two counterintuitive findings — distracted-driver crashes appear *less* fatal (most likely a reporting artifact) and snow/rain crashes are *less* fatal than dry-road crashes (consistent with risk compensation). Our final logistic model achieves AUC = 0.852 on a held-out test set.

1. Dataset

The dataset is the **Crash Data** release from the Virginia Department of Motor Vehicles, published through the Virginia Open Data Portal (Virginia Department of Motor Vehicles 2025). Records are sourced from Virginia's Traffic Records Electronic Data System (TREDS), which centralizes every police-reported crash in the Commonwealth. We downloaded the data on June 11, 2026 from `data.virginia.gov` (URL in the code repository).

Coverage. 379,234 crash records across calendar years 2022 (n = 122,434), 2023 (n = 127,596), and 2024 (n = 129,204), spanning 133 Virginia jurisdictions.

Columns. 74 fields covering crash dynamics (number of vehicles, collision type), conditions (weather, light, road surface, alignment, traffic control), location (lat/long, jurisdiction, urban vs. rural, interstate vs. not), drivers and special factors (alcohol-involved, speed-related, distracted, texting, teen, mature, large truck, work zone, school zone), vulnerable road users (pedestrian, bicycle, motorcycle, moped), and outcomes (`Number_Of_Fatalities`, `Number_Of_Injuries`, plus subgroup-specific counts).

Completeness. Most fields have zero missing values; latitude/longitude is missing for only 5 of 379,234 records. Categorical fields use explicit "Not Provided" or "Unknown" levels rather than blanks.

Data-creation caveats. Records come from police crash reports, which capture only crashes that were investigated by law enforcement. Driver-behavior fields (distraction, texting) are filled in by the reporting officer based on witness statements, scene evidence, and driver interviews — when the driver is killed, these fields are systematically harder to populate, biasing apparent rates downward. We address this in §6.

2. Research question

Primary question. What characteristics of a Virginia crash most strongly predict whether it will be fatal, and how do *raw* (unadjusted) risk factors compare to *adjusted* risk factors after controlling for confounders?

Why this matters. Risk factors that look strongest in descriptive statistics are not necessarily the most important after controlling for context. Knowing the adjusted contribution of each factor helps prioritize interventions (e.g., separated bike lanes vs. cellphone-use enforcement).

Prior work. State DOTs publish annual safety reports based on the same TREDS data (Virginia Department of Motor Vehicles, Highway Safety Office 2024). Academic studies of crash severity using national FARS data have repeatedly identified alcohol, speed, restraint non-use, and vulnerable-user involvement as dominant factors (Savolainen et al. 2011), but typically focus on either fatality-only data (over-representing severe outcomes) or pool many states together. Our

Table 1: Techniques applied to the dataset.

Technique	Yielded a meaningful finding?	New vs. prior work?
Descriptive contingency tables	Yes — set up the story	Repetition (standard ED
Chi-square test (binary + multi-level)	Yes — all top predictors significant	Repetition
Odds ratios with 95% Wald CI	Yes — quantifies effect size	Repetition
Multivariable logistic regression	Yes — separated confounded effects	New (multivariable adju
VIF / GVIF for multicollinearity	Yes — confirmed no problematic collinearity	New
CART (rpart) with loss matrix	Yes — interpretable rule structure	New (loss-weighted tree
Random forest (balanced sample)	Yes — robust importance ranking	New (balanced subsamp

analysis uses a Virginia-only, all-severity dataset, which lets us compute *absolute* risk multipliers for each factor.

3. Research methods

Outcome. We defined a binary `fatal` indicator (`Number_Of_Fatalities` ≥ 1). 2,659 of 379,234 crashes (0.70%) had at least one fatality.

Predictor set. Selected from significant chi-square results; `TypeOfCollision` excluded because its rare categories (Pedestrian, Bicyclist, Motorcyclist) directly mirror the binary involvement flags. The final logistic model uses 16 predictors after dropping rows with “Not Provided” / “Unknown” on key fields (working sample n = 367,195; 80/20 train/test split). The tree models use the same predictors.

Imbalance handling. With a 0.7% positive rate, default rpart splits would never fire. We applied a loss matrix penalising missed-fatal classifications at a ratio of 140:1, matching the class imbalance. For the random forest, we trained on a 50/50 balanced subsample (n = 5,228).

4. Project plan

Meeting cadence. The team met twice weekly via Zoom — Mondays to plan the week’s work and Thursdays to review results. All code was shared through a GitHub repository; members submitted pull requests so that each analysis step could be reviewed before merging. Between meetings, coordination happened through a shared group chat.

Roles. Yousif led data acquisition and exploratory data analysis (scripts `00_download.R` through

`02_eda.R`), including identifying the Virginia Open Data Portal source and confirming data completeness. Hebba led statistical modeling — hypothesis tests, logistic regression, VIF diagnostics, and both tree models (scripts `03_hypothesis_tests.R` through `05_tree.R`). Alex led figure generation, report writing, and references (script `06_figures.R` and `report.Rmd`). All three members reviewed each other's sections and contributed edits to the final write-up.

Timeline.

Week	Dates	Tasks
1	May 26 – May 30	Dataset selection; download and initial schema inspection; research question scoping
2	Jun 2 – Jun 6	EDA (fatality rates by predictor); chi-square hypothesis tests and odds ratios
3	Jun 9 – Jun 10	Logistic regression; VIF diagnostics; CART and random forest models
4	Jun 10 – Jun 11	Figure generation; report writing and knitting; presentation preparation

Skills learned. The project required learning several techniques beyond standard coursework: (1) fitting and interpreting multivariable logistic regression on a large imbalanced dataset, including computing adjusted odds ratios and comparing them to unadjusted estimates; (2) handling severe class imbalance using a loss matrix in CART and balanced subsampling for random forests; (3) diagnosing multicollinearity via generalized VIF (GVIF) for models with multi-level factors; (4) building reproducible R pipelines with modular scripts and version control; and (5) producing a literate statistical report in R Markdown compiled to PDF.

5. Analyze data and results

5.1 Raw fatality rates by risk factor

Figure 1 shows unadjusted odds ratios for each binary risk factor against the population baseline fatality rate of 0.70%. Pedestrian-involved crashes are 15× more likely to be fatal; alcohol-involved crashes 8×.

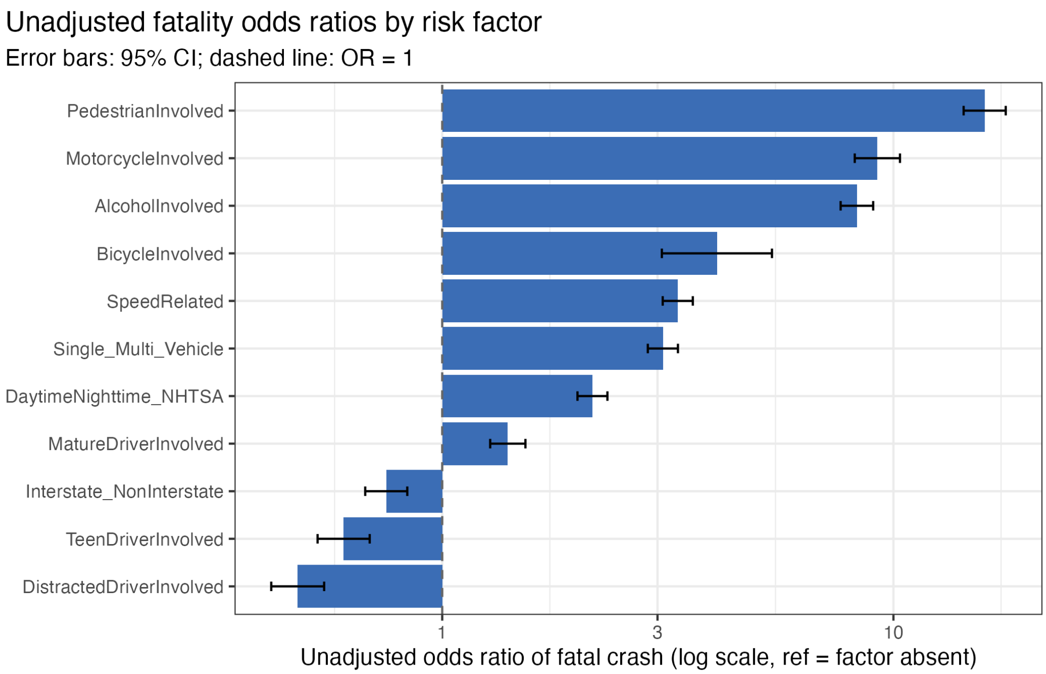

Figure 1: Unadjusted odds ratios of fatal crash for binary risk factors. Bars show OR with 95% CI; dashed line marks OR = 1 (no effect). Log scale.

5.2 Adjusted risk factors (multivariable logistic regression)

Table 2 and Figure 2 report adjusted odds ratios from the multivariable logistic regression.

5.3 Adjustment changes the story

Figure 3 overlays unadjusted vs. adjusted ORs for each binary risk factor. **Pedestrian, bicycle, and mature-driver effects grow stronger** after adjustment; **nighttime and interstate effects wash out** because they were proxies for darkness and rural location respectively.

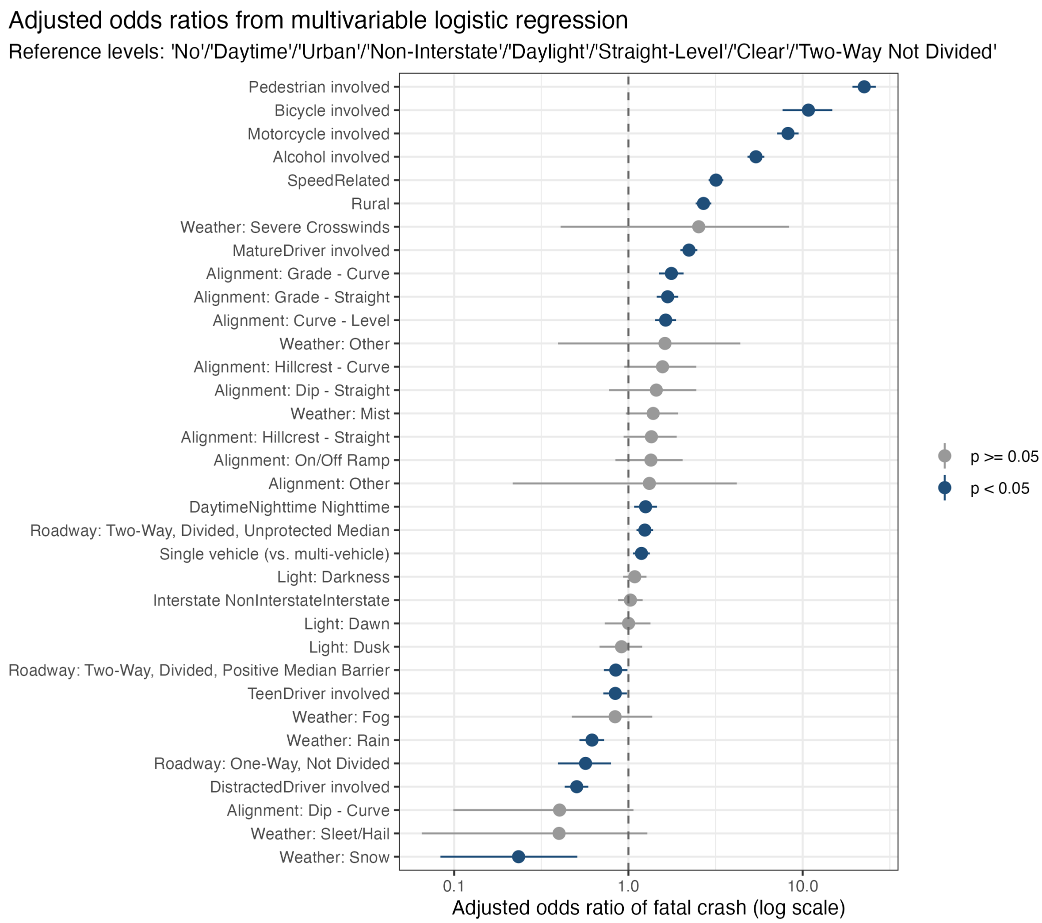

Figure 2: Adjusted odds ratios from multivariable logistic regression. Dark points are significant at $p < 0.05$; grey points are not. Log scale.

Table 2: Significant predictors from multivariable logistic regression ($p < 0.05$).

Term	Adjusted OR (95% CI)	p-value
PedestrianInvolvedYes	22.55 (19.31-26.29)	< 1e-200
BicycleInvolvedYes	10.79 (7.67-14.79)	5.9e-46
MotorcycleInvolvedYes	8.24 (7.13-9.48)	4e-186
AlcoholInvolvedYes	5.39 (4.82-6.02)	3.3e-194
SpeedRelatedYes	3.18 (2.89-3.51)	9e-120
LocationTypeNameRural	2.70 (2.43-2.99)	4e-79
MatureDriverInvolvedYes	2.22 (1.99-2.49)	1e-44
RoadwayAlignmentGrade - Curve	1.76 (1.49-2.07)	1.1e-11
RoadwayAlignmentGrade - Straight	1.68 (1.45-1.93)	9.1e-13
RoadwayAlignmentCurve - Level	1.64 (1.42-1.88)	3.2e-12
DaytimeNighttime__NHTSANighttime	1.25 (1.08-1.46)	0.0032
RoadwayDescriptionTwo-Way, Divided, Unprotected Median	1.24 (1.11-1.39)	0.00011
Single__Multi__VehicleSingle Vehicle Crashes	1.19 (1.06-1.33)	0.0022
RoadwayDescriptionTwo-Way, Divided, Positive Median Barrier	0.85 (0.72-0.99)	0.035
TeenDriverInvolvedYes	0.84 (0.72-0.98)	0.028
WeatherConditionRain	0.62 (0.52-0.72)	6.3e-09
RoadwayDescriptionOne-Way, Not Divided	0.57 (0.39-0.79)	0.0015
DistractedDriverInvolvedYes	0.50 (0.43-0.59)	7.8e-18
WeatherConditionSnow	0.23 (0.08-0.51)	0.0013

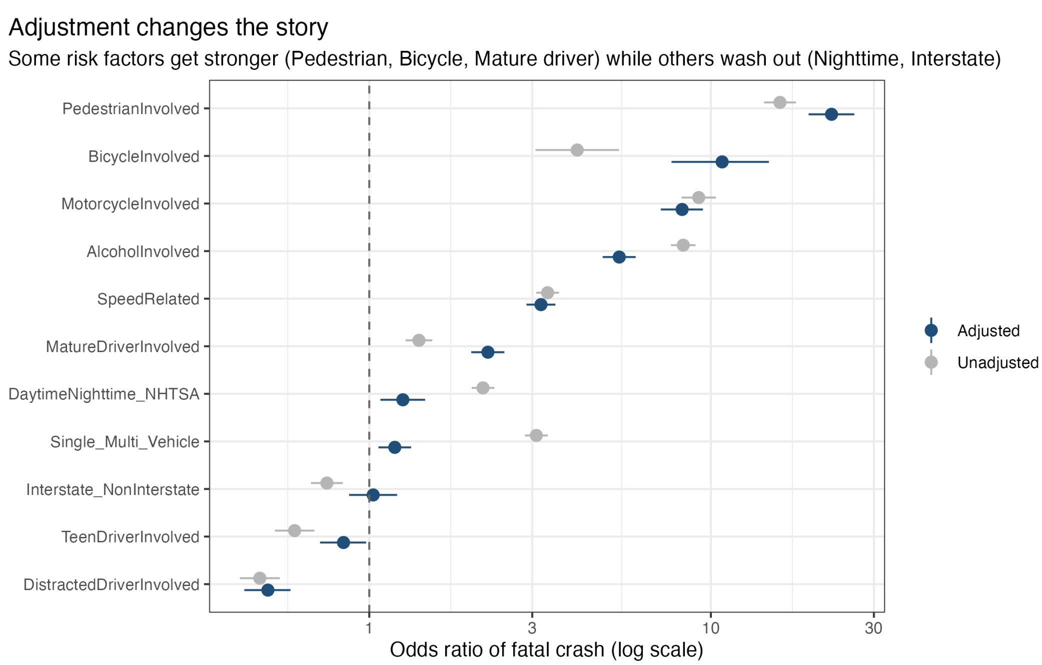

Figure 3: Adjustment changes the story: side-by-side unadjusted vs. adjusted odds ratios.

Table 3: Logistic regression diagnostics.

Metric	Value
n__train	293,756
n__test	73,439
prevalence__train	0.0072
auc__train	0.8563
auc__test	0.8519

5.4 Model diagnostics

VIF / GVIF values were all below 3 (max = 2.93), indicating no problematic multicollinearity.

5.5 Classification tree and ensemble agreement

The CART tree (Figure 4) splits first on `AlcoholInvolved`, then on `PedestrianInvolved` and `SpeedRelated` — mirroring the logistic regression's top coefficients. The random forest variable-importance ranking (Figure 5) corroborates this: alcohol, pedestrian, location, and speed top the list.

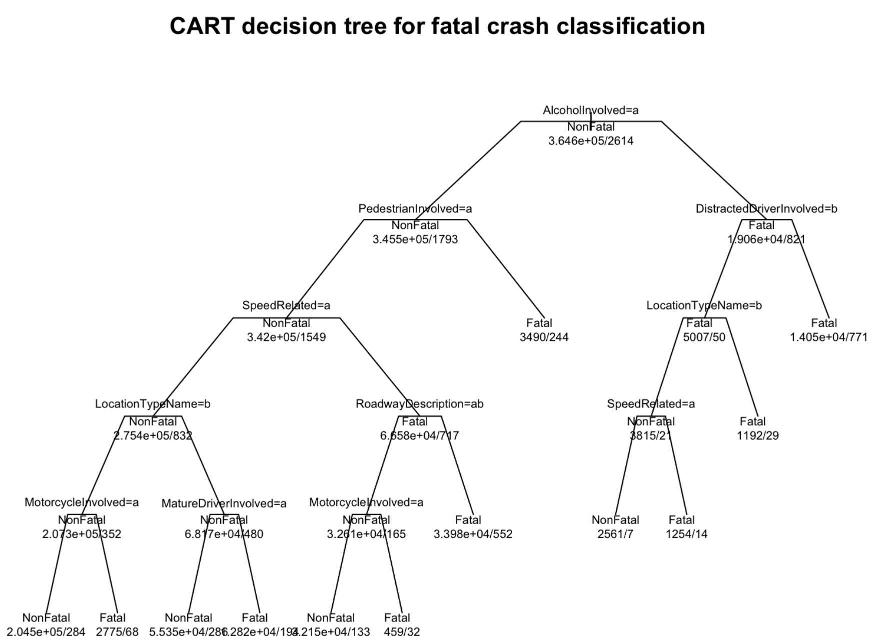

Figure 4: CART decision tree (loss-weighted to address class imbalance). Each terminal node is labeled with the predicted class and (n__NonFatal, n__Fatal) counts.

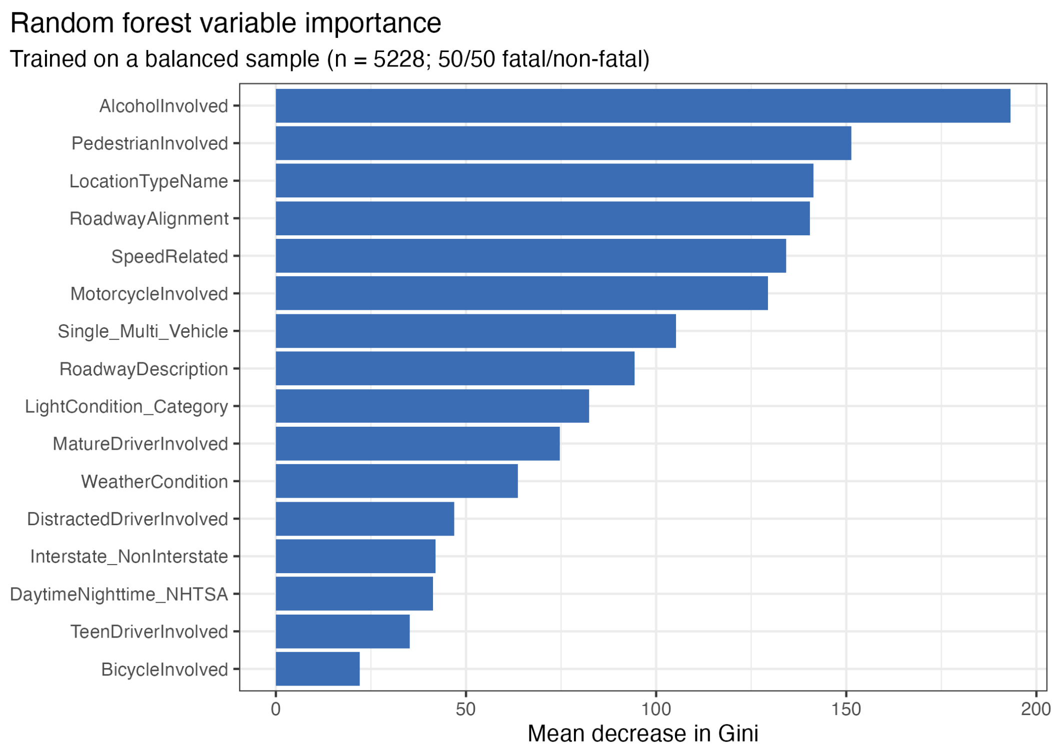

Figure 5: Random forest variable importance (mean decrease in Gini).

5.6 Counterintuitive findings

Adverse weather is less deadly per crash. Snow crashes have an adjusted OR of 0.23 and rain 0.62 — both *below* 1. Drivers appear to compensate for visibly bad conditions by slowing down or staying off the roads.

Distracted-driver crashes appear less fatal (adjusted OR 0.5). Two explanations: (i) reporting bias — when the at-fault driver is killed, no one can confirm distraction; (ii) selection — distracted drivers may be more common in low-speed urban crashes. We cannot fully separate these without survey or telematics data.

6. Summary

Fatal Virginia crashes from 2022–2024 are not a random sample of all crashes; they cluster around a small number of high-risk situations. After adjusting for confounders, the **strongest fatality multipliers** are pedestrian involvement (OR 22.6), bicycle involvement (OR 10.8), motorcycle involvement (OR 8.2), alcohol (OR 5.4), and rural location (OR 2.7). All three modeling methods

— logistic regression, CART, and random forest — agree on this ranking. The model achieves AUC = 0.8519 on held-out data.

Two findings push against intuition. First, **adverse weather is associated with lower per-crash fatality risk**, consistent with the risk-compensation hypothesis: drivers slow down when conditions look obviously dangerous. Second, **distracted-driver crashes are also recorded with lower fatality rates**, but this almost certainly reflects how the field is populated — distraction is hard to confirm post-mortem.

Practical takeaways. Per-crash fatality risk is concentrated in *who and where*, not *when*. Investments in pedestrian infrastructure, motorcycle safety, alcohol-impaired-driving enforcement, and rural-roadway design are likely to yield the largest absolute reductions in fatalities. Conditional-on-crash fatality rates should not be used as the sole metric for evaluating distracted-driving policy — exposure-adjusted analyses are needed.

References

Savolainen, Peter T., Fred L. Mannering, Dominique Lord, and Mohammed A. Quddus. 2011. “The Statistical Analysis of Highway Crash-Injury Severities: A Review and Assessment of Methodological Alternatives.” *Accident Analysis & Prevention* 43 (5): 1666–76. https://doi.org/10.1016/j.aap.2011.03.025.

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